

Safety & Maintenance Manual

REV. 1.0

Boiler & Heating Installations – ThermoMax Pro Series TMX-5000-HE

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APPLIES TO	REVISION
ThermoMax Pro TMX-5000-HE (all units)	1.0 – Initial Release
PREPARED BY	REVIEWED BY
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NEXT REVIEW DATE	DOCUMENT STATUS
July 24, 2027	ACTIVE – Mandatory Compliance

DANGER

This manual must be read in full by all personnel responsible for the operation, maintenance, and inspection of boiler and heating installations. Non-compliance with the safety procedures contained herein may result in serious injury, death, property damage, or voiding of the product warranty.

1. SCOPE & PURPOSE

This Safety and Maintenance Manual establishes the mandatory security measures, operational protocols, and scheduled maintenance procedures for all boiler and heating installations operated by Acme Corporation. It applies specifically to the ThermoMax Pro Series TMX-5000-HE industrial condensing boiler units installed at the Acme Corporation Manufacturing Plant, Hall B, Berlin.

The document is structured as follows: Section 2 covers personnel and access safety. Section 3 addresses operational safety measures. Section 4 details emergency procedures. Section 5 provides the scheduled maintenance programme. Section 6 covers inspection and record-keeping. Section 7 lists applicable regulations and standards.

2. PERSONNEL SAFETY & ACCESS CONTROL

2.1 Authorised Personnel

Only trained, authorised personnel are permitted to operate, inspect, or carry out maintenance on boiler and heating installations. The following roles are defined:

ROLE	AUTHORISED ACTIVITIES	CERTIFICATION REQUIRED
Plant Operator	Day-to-day monitoring, reading gauges, reporting anomalies	In-house boiler operator training
Maintenance Technician	Routine servicing, filter changes, minor adjustments	NVQ Level 2 or equivalent
Certified Engineer	Full service, repair, commissioning, safety valve testing	Gas Safe / DVGW certified
Emergency Responder	Gas isolation, evacuation coordination	Site emergency response training
Manufacturer's Engineer	Warranty repairs, firmware updates, major overhauls	TechSupply GmbH authorised

2.2 Personal Protective Equipment (PPE)

MANDATORY PPE (ALL BOILER ROOM ACTIVITIES)

- Heat-resistant gloves (EN 407) when handling hot components
- Safety footwear (EN ISO 20345, S3) at all times in plant
- Hard hat (EN 397) during installation and overhead work
- Eye protection (EN 166) during inspections and cleaning
- Flame-resistant clothing (EN ISO 11612) near burners

SITUATIONAL PPE (AS ASSESSED)

- Full-face respirator (EN 136) when working with chemicals
- Ear protection (EN 352) in high-noise boiler rooms (>85 dB)
- Self-contained breathing apparatus (SCBA) in confined spaces
- Anti-vibration gloves during prolonged tool use
- High-visibility vest when working near moving plant equipment

2.3 Access Control

- The boiler room must be kept locked at all times. Access is restricted to authorised personnel only. A key register must be maintained.
- A site access log must be completed by all personnel entering the boiler room, recording name, time in, time out, and purpose of visit.
- No lone working is permitted during hot-work, pressure testing, or gas-related maintenance activities. A buddy system must be observed.
- Contractors and third-party engineers must sign in at the site reception and be escorted to the boiler room by an authorised Acme employee.
- Children, unauthorised visitors, and untrained staff must never be permitted access to the boiler room under any circumstances.

3. OPERATIONAL SAFETY MEASURES

3.1 Safe Operating Parameters

PARAMETER	MINIMUM	NORMAL RANGE	MAXIMUM	ACTION IF EXCEEDED
System Pressure	0.5 bar	1.0 – 2.0 bar	3.0 bar	Shut down; check expansion vessel
Flow Temperature	40 degC	60 – 80 degC	85 degC	Reduce setpoint; check thermostat
Return Temperature	30 degC	45 – 65 degC	–	Check circulation pump & valves
Flue Gas Temperature	–	120 – 160 degC	200 degC	Inspect heat exchanger for fouling
CO in Flue Gas	–	<100 ppm	200 ppm	Stop burner; call certified engineer
Gas Supply Pressure	15 mbar	18 – 22 mbar	25 mbar	Contact gas network operator
Combustion Air Flow	–	per burner spec	–	Check air inlet filters; call engineer

3.2 Pre-Start Safety Checks

The following checks must be completed before starting the boiler after any shutdown or maintenance period:

#	CHECK ITEM	METHOD	PASS CRITERIA
1	System water pressure	Read pressure gauge	0.5 – 3.0 bar
2	Gas isolation valve – OPEN	Visual inspection	Valve handle parallel to pipe
3	No gas odour detected	Olfactory / gas detector	No odour / 0 ppm reading
4	Condensate drain – clear and unrestricted	Visual inspection	No blockage; drain free-flowing
5	Air intake and flue outlet – unobstructed	Visual inspection	Clear of debris, ice, or nesting
6	Safety pressure relief valve – not locked or gagged	Visual inspection	Free to operate; seal intact
7	Expansion vessel pre-charge pressure	Gauge check	1.0 bar (cold system)
8	Control panel – no active fault codes	Display check	No fault codes displayed
9	Low water cut-off device – operational	Test function	Burner shuts off as expected
10	All pipe joints and fittings – no visible leaks	Visual inspection	Dry; no staining or corrosion

CAUTION

Never attempt to start the boiler if a gas leak is suspected. Evacuate the area, isolate the gas supply at the external meter, ventilate the room, and contact the gas emergency line (0800 111 1999 in Germany) before re-entering.

3.3 Shutdown Procedure

Follow this sequence for planned shutdowns. For emergency shutdowns, proceed to Section 4.

1

Reduce Load Setpoint

- Lower flow temperature setpoint gradually to 40 degC using the control panel.
- Allow the boiler to complete its current firing cycle before proceeding.

2

Disable Burner

- Set the boiler to standby mode via the control panel.
- Confirm the burner flame has extinguished (check display status).

3

Close Gas Valve

- Close the manual gas isolation valve located on the gas supply pipework.
- Tag the valve with a 'DO NOT OPERATE' label if shutdown is for maintenance.

4

Allow Cool-Down

- Allow the system to cool naturally. Do not force-cool with cold water.
- Monitor system pressure during cool-down; it should decrease steadily.

5

Isolate Electrically

- Switch off the boiler isolator switch.
- For maintenance work, lock out / tag out (LOTO) the electrical supply.

6

Record Shutdown

- Enter shutdown date, time, reason, and operative name in the Plant Log Book.
- Note any fault codes displayed prior to shutdown for engineering follow-up.

4. EMERGENCY PROCEDURES**4.1 Gas Leak****DANGER**

If a gas leak is detected (smell of gas, CO detector alarm, or hissing sound from pipework): DO NOT operate any electrical switches. DO NOT use mobile phones in the area. Evacuate all personnel immediately.

- 1 Shout 'GAS LEAK – EVACUATE' to alert all personnel in the boiler room and adjacent areas.
- 2 Activate the nearest fire alarm pull station to initiate site evacuation.
- 3 Close the external gas isolation valve (located at the meter cabinet outside the plant building) if safe to do so.
- 4 Do NOT switch any electrical equipment on or off. Do not use light switches or any ignition source.
- 5 Assemble at the designated muster point (Car Park B, north side of the building).
- 6 Call the gas emergency service: 0800 111 1999 (24h, Germany) and site emergency: +49 30 1234 9999.
- 7 Do not re-enter the building until the gas network operator and emergency services have confirmed it is safe.

4.2 Boiler Over-Pressure Event**CAUTION**

If the system pressure exceeds 3.0 bar or the pressure relief valve activates:

- 1 Do not attempt to manually reseal or block the pressure relief valve discharge.
- 2 Switch the boiler to standby mode using the control panel immediately.
- 3 Allow the system to cool down; monitor pressure gauge – it should fall as the system cools.
- 4 Do not restart the boiler until the cause of over-pressure has been identified and rectified.
- 5 Contact TechSupply GmbH warranty/service line: +49 89 9876 5432. Log the event in the Plant Log Book.
- 6 If pressure does not fall or the PRV continues to discharge, evacuate and call emergency services.

4.3 Carbon Monoxide (CO) Alarm**DANGER**

CO is odourless and lethal. If the CO alarm activates (alarm threshold: 50 ppm):

- 1 Evacuate the boiler room and all adjacent areas immediately. Do not delay to investigate.
- 2 Close the manual gas isolation valve at the meter cabinet outside if safe to do so.
- 3 Call the emergency services (112) and gas emergency line (0800 111 1999).
- 4 Seek fresh air immediately. If any person has symptoms (headache, dizziness, nausea), call ambulance (112).

- 5 Do not re-enter until fire service personnel confirm CO levels are below 10 ppm.
- 6 A full combustion analysis must be performed by a certified engineer before the boiler is restarted.

4.4 Fire in the Boiler Room

- 1 Activate the nearest fire alarm and call 112 immediately.
- 2 Attempt to fight only a very small, contained fire using a CO2 extinguisher (never water on a gas flame).
- 3 If the fire cannot be immediately controlled, evacuate. Do not put yourself at risk.
- 4 Close the external gas isolation valve if safe to access without passing through the fire.
- 5 Assemble at the designated muster point and await emergency services.
- 6 Do not re-enter until fire services confirm the area is safe and all ignition sources are secured.

4.5 Emergency Contact Directory

SERVICE	NUMBER	AVAILABILITY
Emergency Services (EU)	112	24 / 7
Gas Emergency – Germany	0800 111 1999	24 / 7
Site Emergency Coordinator	+49 30 1234 9999	24 / 7
TechSupply GmbH Service Line	+49 89 9876 5432	Mon–Fri 07:00–18:00
TechSupply GmbH Emergency	+49 89 9876 0000	24 / 7
Acme Facilities Manager	+49 30 1234 5010	Mon–Fri 08:00–17:00
Poison Control Centre	030 19240	24 / 7

5. SCHEDULED MAINTENANCE PROGRAMME

All maintenance activities must be carried out by personnel with the appropriate certification level (see Section 2.1). Activities requiring a Certified Engineer must be performed by TechSupply GmbH or an approved third-party contractor holding valid Gas Safe / DVGW certification. All completed work must be logged in the Plant Log Book.

5.1 Daily Operator Checks

CHECK ITEM	METHOD	RECORD
System pressure reading (target 1.0–2.0 bar)	Read gauge	Log Book
Flow and return temperature readings	Read display	Log Book
No active fault codes on control panel	Visual check	Note any faults
No unusual noises (knocking, hissing, rattling)	Auditory check	Log Book
CO alarm functional indicator light – GREEN	Visual check	Log Book
No visible water leaks or condensate overflow	Visual inspection	Log Book
Gas meter reading	Read meter	Weekly total

5.2 Monthly Maintenance Tasks

TASK	DURATION	BY
Test low water cut-off device – confirm burner shutdown	15 min	Maint. Tech.
Test pressure relief valve – check for correct lift and re-seating	10 min	Maint. Tech.
Inspect condensate trap and neutralisation cartridge – replace if needed	20 min	Maint. Tech.
Check expansion vessel pressure (cold system, 1.0 bar target)	15 min	Maint. Tech.
Inspect all visible pipework joints and fittings for leaks	20 min	Maint. Tech.
Verify BMS integration – confirm setpoints and alarm thresholds correct	15 min	Plant Operator
Clean control panel and check cable integrity	10 min	Plant Operator

5.3 Annual Service – Certified Engineer (Full Service)

SERVICE ITEM	STANDARD / TARGET
Combustion analysis – CO, CO ₂ , O ₂ , flue temperature	CO <100 ppm; O ₂ 3–5%; efficiency >94%
Full heat exchanger inspection and chemical descaling	Clear of scale and fouling
Burner nozzle and ignition electrode inspection / replacement	Per manufacturer spec
Gas valve leak test and seat integrity check	Zero leakage confirmed
Safety thermostat and high-limit thermostat calibration test	Cuts out at set temperature
Flue integrity pressure test	No leakage at 20 Pa
System water quality analysis: pH, hardness, inhibitor concentration	pH 7.5–9.5; inhibitor 300–500 ppm
Anti-corrosion inhibitor top-up if below specification	Sentinel X100 or equivalent
Pump operation, head pressure, and seal condition check	Full flow; no seal leak
All safety devices tested – PRV, LWCO, overheat stat	All activate correctly
Full controls and BMS interlock test	All sequences confirmed
Service completion report issued and countersigned	Warranty record updated

NOTE

The annual full service must be booked with TechSupply GmbH at least 30 days in advance to guarantee engineer availability and preserve extended warranty coverage on the heat exchanger and burner assembly. Contact: warranty@techsupply.de or +49 89 9876 5432.

6. INSPECTION, RECORDS & COMPLIANCE

6.1 Plant Log Book

A physical Plant Log Book must be kept in the boiler room at all times. The following must be recorded:

- Daily operator checks – date, time, readings, operative name and signature
- Any faults, alarms, or anomalies observed – including fault codes, corrective action taken, and outcome
- All maintenance activities – date, scope, operative name, certification number, and outcome

- All access entries – every person entering the boiler room, with times and purpose
- Any changes to operating setpoints, BMS configuration, or safety device settings
- All warranty claim events – date, fault description, TechSupply reference number

6.2 Statutory Inspections

INSPECTION TYPE	FREQUENCY	GOVERNING REGULATION	INSPECTOR
Pressure vessel examination (BG-Check)	Annually	BetrSichV / PED 2014/68/EU	Notified Body (TUV or equiv.)
Gas installation safety check	Annually	DVGW-TRGI 2018 / GasHizV	Gas Safe / DVGW Engineer
Electrical installation inspection	Every 3 yrs	DGUV V3 / DIN VDE 0105	Qualified Electrician
Flue and chimney inspection	Annually	1. BImSchV / EN 15287	Chimney Sweep (Schornsteinfeger)
CO alarm calibration and test	Annually	EN 50291 / Manufacturer spec	Certified Technician
Fire suppression / extinguisher check	Annually	DIN 14406 / BetrSichV	Fire Equipment Specialist

CAUTION

Failure to comply with statutory inspection requirements may result in enforcement action by the Gewerbeaufsichtsamt (Trade Supervisory Authority), invalidation of operating licences, and full voiding of the product warranty. Inspection certificates must be retained for a minimum of 10 years.

7. APPLICABLE REGULATIONS & STANDARDS

This manual has been prepared in accordance with the following German, EU, and international regulations, directives, and technical standards. All operations must comply with the most current version of each:

REFERENCE	TITLE	SCOPE
BetrSichV	Betriebssicherheitsverordnung – German Industrial Safety Regs.	Pressure systems, safe operation
GasHizV	Gasheizungsverordnung – Gas Heating Appliance Ordinance	Gas-fired heating equipment
DVGW-TRGI 2018	Technical Rules for Gas Installations	Gas pipework, installation, testing
PED 2014/68/EU	Pressure Equipment Directive	Boiler pressure vessel classification
EN 303-5	Heating Boilers – Part 5: Solid Fuel Boilers	Construction and testing
EN 15502	Gas-fired central heating boilers	Requirements and test methods
EN 50291	Gas Detectors for Domestic Use	CO alarm requirements
1. BImSchV	Federal Immission Control Ordinance	Flue gas emissions limits
DGUV V3	Electrical Installations and Equipment	Electrical safety
DIN EN ISO 11612	Protective Clothing against Heat and Flame	PPE standard
REACH Regulation	EC 1907/2006 – Chemical Safety	Water treatment chemicals

8. DOCUMENT APPROVAL & DISTRIBUTION

This Safety and Maintenance Manual has been reviewed and approved by the parties below. All personnel listed in the distribution list have received and acknowledged this document. A copy must remain posted in the boiler room at all times.

PREPARED BY	REVIEWED BY	APPROVED BY
Victor Clar	TechSupply GmbH Engineering	Acme Corp. Facilities Director
Procurement Manager	Certified Engineers Team	
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

NOTE

This document must be reviewed annually or whenever significant changes are made to the installation, operating conditions, or applicable regulations. The next scheduled review date is July 24, 2027. Document owner: Victor Clar, victor.clar@spryker.com.